

MODEL QUESTION PAPER

FLUID MECHANICS

Max.Marks : 75

Part A

I. Answer all questions in one word or one sentence. Each question carries 1 mark

9 X 1 = 9 Marks

1. State Newton's law of viscosity
2. State Pascal' law
3. The velocity profile for turbulent flow through a closed conduit is.
4. The friction factor for turbulent flow in a hydraulically smooth pipe depends only on
5. Define NPSH
6. Identify the pump used for draining of shallow pits
7. List types of impeller in Centrifugal pump
8. Define Schedule number
9. Nominal size of the discharge pipe of a pump is usually _____ the nominal size of the inlet pipe

Part B

II. Answer any EIGHT questions from the following. Each questions carries 3 marks

8 X 3 = 24 Marks

1. Discuss the classification of fluids based on Newton's law of viscosity
2. List any 3 desirable properties of manometric fluid
3. Differentiate viscous and turbulent flow
4. State and explain Bernoullis theorem.
5. Derive continuity equation for flow though a closed conduit
6. Draw the diagram of a centrifugal pump and label its parts
7. List the standard operating procedure to start a centrifugal pump
8. Summarise the use of dampner in positive displacement pump.
9. Discuss the color coding of pipes in industry
10. Draw the layout of butterfly valve and label its parts

Part C

Answer ALL questions. Each questions carries 7 marks

6 X 7 = 42 Marks

III. Illustrate the working of U tube manometer.

OR

IV. Illustrate the working of inclined tube manometer

V. Summarise Reynolds Experiment

OR

VI. Derrive Bernoullis Equation

VII. Summarise losses due to friction in pipe line

OR

VIII. A liquid with density of 110kg/m^3 flows through a straight pipe having inside diameter 5cm at the rate of 45kg/hr . The viscosity of liquid is 0.9 centi poise. Calculate the pressure dorp due to friction in the pipe, if pipe line is 1.6km long.

IX. Discuss the safety precautions to be taken to start a positive displacement pump..

OR

X. Explain the working of a multistage compressor

XI. Illustrate steam ejector

OR

XII. Illustrate the pump characteristics of a centrifugal pump

XIII. Summarise the working of a Gate valve

OR

XIV. Summarise the working of a ventury meter

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Course : Fluid Mechanics

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Mark Distribution

Module	Hr/Module	Marks / Module	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	12	25	2	2	2	6	2	14	6	22
2	18	37	2	2	3	9	4	28	9	39
3	18	37	2	2	3	9	4	28	9	39
4	12	25	3	3	2	6	2	14	7	23
Total	60	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	36	29
Understanding	59	48
Applying	28	23
Analysing	-	
Evaluating	-	
Creating	-	
Total	123	100

Question Wise Analysis

Course : Fluid Mechanics

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M 1.02	R	1	
I.2	M 1.03	R	1	
I.3	M 2.01	R	1	
I.4	M 2.04	R	1	
I.5	M 3.01	R	1	
I.6	M 3.01	R	1	
I.7	M 3.01	R	1	
I.8	M 4.01	R	1	
I.9	M 4.01	R	1	
II.1	M 1.02	R	3	
II.2	M 1.04	R	3	
II.3	M 2.01	R	3	
II.4	M 2.02	R	3	
II.5	M 2.01	U	3	
II.6	M 3.01	R	3	
II.7	M 3.01	R	3	

II.8	M 3.02	R	3	
II.9	M 4.01	R	3	
II.10	M 4.03	R	3	
III.	M 1.04	U	7	
IV.	M 1.04	U	7	
V.	M 2.01	U	7	
VI.	M 2.02	U	7	
VII.	M 2.04	U	7	
VIII.	M 2.02	A	7	
IX.	M 3.02	U	7	
X.	M 3.04	U	7	
XI.	M 3.04	A	7	
XII.	M 3.04	A	7	
XIII.	M 4.03	U	7	
XIII.	M 4.04	U	7	